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Low-Complexity Geometry-Aware Hybrid Beamforming for Near-Field STAR-RIS Networks

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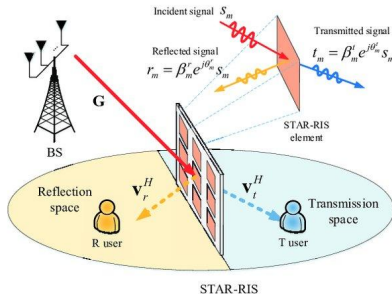
1. Context & Motivation

STAR-RIS: A simultaneously transmitting and reflecting surface that covers both sides of space. In energy-splitting mode each element sends part of the energy through and reflects the rest. Users only receive; the T-side or R-side is decided by location.

Hybrid beamforming: The base station uses a limited number of RF chains. Analog phase shifters plus a small digital precoder keep hardware cost low, so only a few users can be scheduled in each slot.

Constraint: One common profile per side couples user scheduling, energy splitting, and beamforming. Iterative joint optimization grows fast with the surface size, and gain-only scheduling keeps favoring a few users whose effective channels can still be highly correlated.

2. Overall Architecture



STAR-RIS architecture

Hybrid stream budget. $|S| \leq N_s \leq N_{\text{RF}}$, per-side caps $N_T^{\text{max}}, N_R^{\text{max}}$; one common v_T, v_R per slot with $\beta_T + \beta_R = 1$. The design returns $S, v_T, v_R, F = F_{\text{RF}} F_{\text{BB}}$.

Near field. Large surface at 28 GHz: a beam depends on angle *and* distance.

$$[G]_{n,m} = \frac{1}{\sqrt{N_t} d_{B,m \rightarrow n}^{\eta_{BR}/2}} e^{-j \frac{2\pi}{\lambda} d_{B,m \rightarrow n}}, \quad [r_k]_n = d_{n \rightarrow k}^{-\eta_{RU}/2} e^{-j \frac{2\pi}{\lambda} d_{n \rightarrow k}}$$

cascaded channel $h_k^H = r_k^T \text{diag}(v_k) G$.

3. Three Ingredients Aligned in $\Psi(S)$

Proposed GA-HBF-PFBDP — three geometry scores, one subset utility

$$\Psi(S) = \sum_{k \in S} \psi_k \log\left(1 + \frac{\hat{g}_k}{\sigma^2 + \mu l_k(S)}\right), \quad l_k(S) = \sum_{i \in S \setminus \{k\}} \rho_{k,i}^2 \hat{g}_i$$

computed before any ν , β , or F exists (16)

(1) Link strength

$$\hat{G}_k = \sum_{n=1}^N |[r_k]_n|^2.$$

$$\frac{1}{N_t} \sum_{m=1}^{N_t} |[G]_{n,m}|^2$$

Cascaded BS-RIS-user power over the aperture.

Independent of ν , β .

(2) Fairness

$$\psi_k = \frac{\hat{G}_k}{T_k + \epsilon}$$

T_k : user's long-term rate.
Strong users first, but not the same ones every slot.

(3) Correlation

$$\rho_{i,j} = \frac{|\tilde{h}_i^H \tilde{h}_j|}{\|\tilde{h}_i\|_2 \|\tilde{h}_j\|_2 + \epsilon_c}$$

How aligned two effective channels are; penalized through $l_k(S)$.

4. Who Is Served (S) and How One Surface Serves Both Sides

PF-BDP — exact subset search despite non-additive Ψ

$$\max_S \Psi(S) \text{ s.t. } |S| \leq N_s, |S \cap \mathcal{K}_s| \leq N_s^{\max}; \quad I(m \cup \{k\}) = I(m) + \hat{g}_k [\rho_{1,k}^2, \dots, \rho_{L_c,k}^2]^T$$

Bitmask state

Mask m = users chosen from a pool of L_c .

Accumulator

$I(m)$ carries all pair coupling; $O(L_c)$ update.

Exact

Matches exhaustive to 10^{-14} at $O(2^{L_c} L_c)$.

Energy split and common profiles — one budget, one profile per side

$$D_s = \sum_{k \in S \cap \mathcal{K}_s} \hat{G}_k, \quad \beta_T = \frac{D_T}{D_T + D_R + \varepsilon_\beta}, \quad \beta_R = 1 - \beta_T, \quad v_s = \sqrt{\beta_s} e^{-j\angle(a_s)}$$

Side demand

$$D_s = \sum_{k \in S \cap \mathcal{K}_s} \hat{G}_k$$

stronger side asks more

Energy split

$$\beta_T = \frac{D_T}{D_T + D_R + \varepsilon_\beta}$$

$\beta_R = 1 - \beta_T$, floor-clipped

Common profile

$$v_s = \sqrt{\beta_s} e^{-j\angle(a_s)}$$

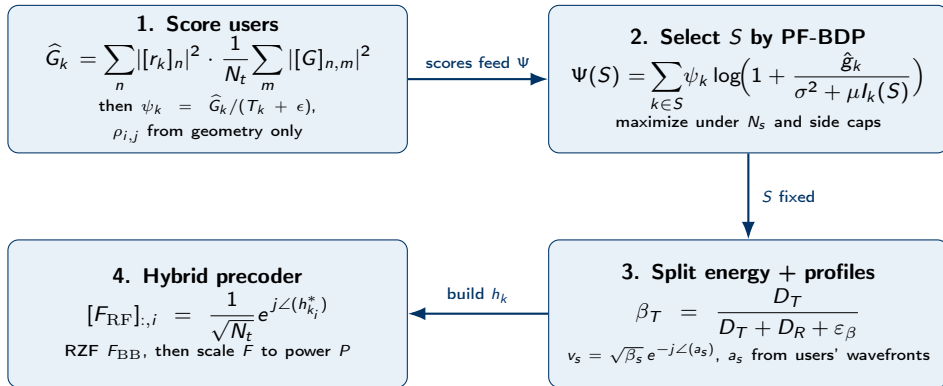
a_s : users' wavefronts, weight $\propto \hat{G}_k$

5. GA-HBF-PFBDP Pipeline

I/O. G BS-RIS channel, r_k RIS-user channel, T_k user's long-

term rate, N_s stream budget, $N_s^{\max}$ per-side caps, P transmit power.

Output: S served users, v_T, v_R common profiles, F hybrid precoder. No alternating optimization.



Per-slot cost $O(NN_t + KN + L_c^2 N_t + 2^{L_c} L_c + N_s N)$: only the bitmask term is exponential, in L_c not N .

6. Simulation Results (Rate)

Rate. Best among practical hybrids; **81–87%** of iterative joint NF STAR-RIS AO at **27–87 $\times$** lower run-time, close to the genie FD-NF-SVD-UB.

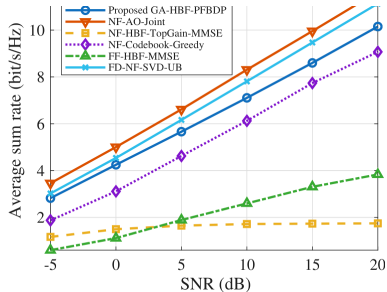


Fig. 1. Average sum rate vs. SNR (300 drops).

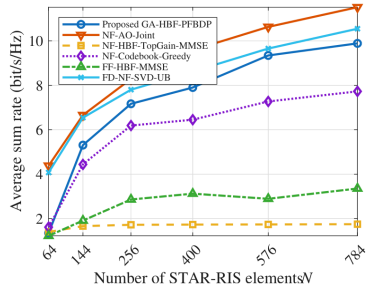


Fig. 2. Average sum rate vs. STAR-RIS elements (300 drops).

7. Simulation Results (Runtime and Users)

Takeaways. (1) Up to $3.5\times$ lower design runtime than codebook search; scales to $N=2304$. (2) Closed-form split within 0.8% of a 29-point exhaustive search (7.05 vs. 7.10 bit/s/Hz). (3) PF-BDP beats top-gain and stays robust as K grows.

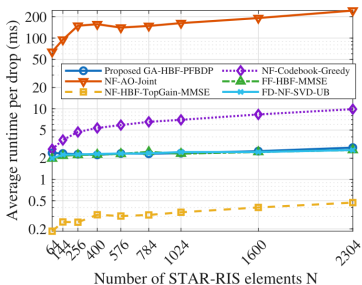


Fig. 3. Average design runtime per drop vs. STAR-RIS elements.

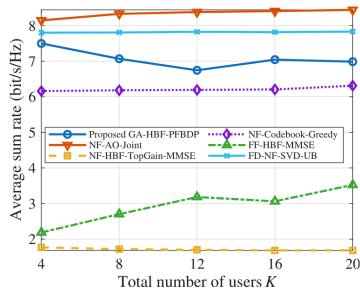


Fig. 4. Average sum rate vs. total users (300 drops).

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Thank You

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